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Superseding
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MILITARY SPECIFICATION

INDICATOR, PRESSURE, FUEL, 0-600 PSI

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of hermetically sealed, remote indicating, synchro-style, fuel pressure indicator.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 The following publications, of the issue in effect on date of invitation for bids, shall form a part of this specification to the extent specified herein:

2.1.1 Specifications.-

Military

MIL-I-7057	Indicators, Synchro, Aircraft, General Specification for
MIL-T-6302	Transmitter, Pressure, Fuel, 0-600 PSI

2.1.2 Drawings.-

Military Standards (Sheet)

MS28005	Transmitters, Pressure - Synchro Style, 320-Degree Movement
MS28010	Indicators, Pressure - Synchro, Single, 2-Inch Size

Air Force-Navy Aeronautical Standard Drawing

AND10404	Pointers - Standard Design of Aircraft Instrument

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The requirements specified in Specification MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein.

3.1.1 In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

3.2 Design.- The indicator shall be designed for use with an MS28005-7 fuel pressure transmitter in accordance with Specification MIL-T-6302.

3.3 Construction.-

3.3.1 Case.- The case shall be hermetically sealed and the case dimensions shall conform to Standard MS28010.

3.3.1.1 A separate nonhermetically sealed cap may be used on the back of the case for the purpose of installing the electrical receptacle. The cap shall be made of nonferrous, low-density metal.

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3.3.1.2 The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.3.1.3 Case Finish.- The portion of the case visible when mounted in the normal instrument panel shall be finished in durable dull black.

3.3.2 Pointer.- The pointer shall be in accordance with AND10404-8, except that the length of the pointer shall be such that the tip will extend into the scale a distance equal to 1/3 to 2/3 of the length of the shortest graduation.

3.3.2.1 Pointer Finish.- The shaded portion of the pointer shall be finished with fluorescent-luminescent material.

3.3.3 Dial.- The dial shall be as shown on figure 1. A minimum diameter of 1-3/4 inches shall be maintained across the outside ends of the graduations. The graduation for 300 psi shall be located on the horizontal center line at the left side of the dial. The electrical zero reference mark shall be located 160 degrees clockwise from the zero graduation.

3.3.3.1 Dial Markings.-

3.3.3.1.1 Fluorescent-Luminescent Markings.- The markings shall be finished in fluorescent-luminescent material. The dimensions of the markings shall be as follows:

	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
Numerals 0, 50, 100, 300, and 500	5/32	0.025
Graduations at 0, 50, and each 100	3/16	0.031
Each 10-pound graduation 10 to 90	3/16	0.020
Minor graduations	3/32	0.020
Lettering "FUEL PRESS"	1/8	0.025
Lettering "PSI"	1/8	0.020

3.3.3.1.2 Durable Dull Black Markings.- The markings MS28005-7 shall be permanently and legibly marked on the dial in 1/16-inch letters. These markings and all other markings not otherwise specified shall be finished in durable dull black.

3.3.4 Weight.- The weight of the indicator shall not exceed 0.60 pound.

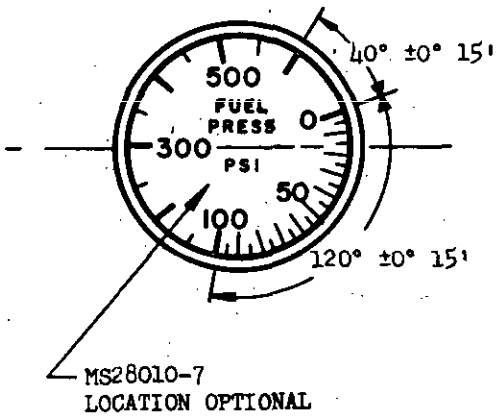
3.4 Identification of Product.- The nameplate shall be marked with the following information:

INDICATOR, PRESSURE, FUEL, 0-600 PSI
MS Part No.
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name or Trade-Mark
U S Property

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 General.- The fuel pressure indicator shall be subjected to inspection and tests in accordance with Specification MIL-I-7057.

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DIMENSIONS IN INCHES.

FIGURE 1. Dial

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4.2 Test Tolerances.- The indicator shall be within the tolerances specified in table I and II of this specification when tested in accordance with Specification MIL-I-7057.

TABLE I

Pressure Indicator Tolerances ±psi			
Test psi	Room Temperature	High Temperature	Low Temperature
0	1.0	1.0	1.0
50	1.0	1.0	1.0
100	1.0	1.0	1.0
150	3.0	3.0	4.0
200	3.0	3.0	4.0
300	3.0	3.0	4.0
400	3.0	3.0	4.0
500	3.0	3.0	4.0
600	3.0	3.0	4.0

TABLE II

Test	Tolerance psi	
	0-100 psi	Over 100 psi
Position Error at 200 psi	1.0	2.0
Friction		
Room Temperature	1.0	6.0
High Temperature	1.0	6.0
Low Temperature	1.5	9.0
Vibration		
Pointer Oscillation	0.5	3.0
Pointer Variation	0.5	3.0

5. PREPARATION FOR DELIVERY

5.1 General.- The fuel pressure indicator shall be packaged, packed, and marked for shipment in accordance with Specification MIL-I-7057, except as specified herein.

5.2 Package Marking.- Each interior package shall be marked with the following information:

INDICATOR, PRESSURE, FUEL, SYNCHRO, 2-INCH SIZE, 0-600 PSI
26 Volts 400 Cycles
MS Part No.
Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Date of Manufacture
Quantity
Name of Manufacturer
Name of Contractor (if different from manufacturer)

IF IN STORAGE AFTER (DATE)*, THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE ISSUE.

* This date shall be 36 months from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

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6. NOTES

6.1 Intended Use.- The fuel pressure indicator covered by this specification is intended for use in conjunction with a suitable transmitter in indicating remotely the pressure in the fuel systems of aircraft engines.

6.2 General.- The instructions contained in Section 6 of Specification MIL-I-7057 are applicable to and form a part of this specification.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Air Force

Other interest:
Navy - A

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